



Real-Time and Embedded Systems Introduction



Luís Miguel Pinho
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General Information

▶ Real-time and Embedded Systems

- ▶ This course addresses the subject of Real-time and Embedded Systems (RTES). It covers the basic concepts and principles of real-time systems such as scheduling algorithms and resource sharing mechanisms as well as criticality levels. The underlying system domain are the embedded systems.

▶ Specific objectives

- ▶ Evaluate concepts and principles associated with real-time systems
- ▶ Describe embedded computing systems
- ▶ Evaluate scheduling algorithms and schedulability analysis for uniprocessor systems;
- ▶ Evaluate mechanisms for shared resources management in RTES;
- ▶ Evaluate principles and techniques for multiprocessor and mixed/multi-criticality in RTES.

General Information

▶ Prerequisites

- ▶ This course assumes students have previously acquired knowledge and skills in:
 - ▶ Algorithms
 - ▶ The principles of computer technologyMethodology

▶ Lectures and laboratory classes have been designed as complementary

- ▶ But schedule is not built on lecture -> lab
- ▶ Students should take both types of classes
 - ▶ Students with special status (e.g., working students) only mandatory to take labs
 - it is expected that students study autonomously the concepts and techniques which are provided in the lectures

Program

- ▶ CP1. Real-time and embedded computing systems
- ▶ CP2. Scheduling algorithms and schedulability analysis for uniprocessor systems
- ▶ CP3. Mechanisms for shared resources management in RTES
- ▶ CP4. Principles and techniques for multiprocessor and mixed/multi-criticality in RTES

Class schedule

- ▶ **Usually**
 - ▶ Lectures on Tuesday/Thursday 18:10 – 20:00
 - ▶ Labs on Thursday 20:40 – 23:30
- ▶ **However**
 - ▶ Changes may occur, will be informed beforehand

Technologies and tools

▶ Technologies and tools to use

- ▶ For most of the course Linux will be used as the underlying “RTOS”
 - ▶ Although not an RTOS, it provides support to the POSIX real-time specification, as well as CBS scheduling
 - ▶ And provides relevant tools for tracing execution
- ▶ Programming
 - ▶ C programming in a Linux environment
- ▶ Two analysis tools will be used
 - ▶ SimSo: <https://projects.laas.fr/simso/>
 - ▶ Cheddar: <https://beru.univ-brest.fr/cheddar/>

Bibliography

▶ Main

- ▶ Teaching Slides and laboratory guides
- ▶ Giorgio C. Buttazzo. "Hard Real-Time Computing Systems: Predictable Scheduling Algorithms and Applications (3rd ed.)". Springer Publishing Company, 2011.

▶ Other

- ▶ Material and links provided in Moodle
- ▶ Robert I. Davis and Alan Burns. "A survey of hard real-time scheduling for multiprocessor systems". ACM Comput. Surv. 43, 4, Article 35 (October 2011).
- ▶ Alan Burns and Robert I. Davis. "A Survey of Research into Mixed Criticality Systems". ACM Comput. Surv. 50, 6, Article 82 (November 2017).
- ▶ Alan Burns and Andy Wellings. "Real-Time Systems and Programming Languages (Fourth Edition)". Addison Wesley Longman, 2009.

Evaluation

- ▶ **Evaluation during the term**
 - ▶ Laboratory work in the context of PL classes
 - ▶ To be presented in PL class
 - ▶ Deadlines are soft real-time
 - 48 hours of delay allowed
 - 15% decrease on grade after each 24 hours
 - After 48 hours implies grade of 0
 - ▶ The grade is individual (evaluation includes individual assessment during the lab)
 - ▶ Quiz in the context of T class
 - ▶ Small problem to be solved in one class
 - ▶ If student is not in the class, the quiz can be retaken in the exam
- ▶ **Exam after the term**
 - ▶ Can include retaking quiz
- ▶ **Grade improvement**
 - ▶ Exam for total grade

Ethics

- ▶ The curricular unit follows the general code of conduct of the Polytechnic Institute of Porto
- ▶ Students are encouraged to discuss the topics of the classes with their peers, but the developed code and reports need to be individual work of the students (or in predefined groups)
 - ▶ all uses of external items need to be checked with the professor
 - ▶ and properly credited
- ▶ Any violation of this code by a student implies that the specific evaluation item is nulled
 - ▶ and the violation is reported
- ▶ The use of AI tools is allowed as a learning tool
 - ▶ Can be used for brainstorming, explain, or code
 - ▶ But it is expected that students used it to learn, evaluations will assess student knowledge, not produced code.

Teaching Staff

- ▶ **Responsible**

- ▶ Luís Miguel Pinho (LMP@isep.ipp.pt)

- ▶ **Lectures**

- ▶ Luís Miguel Pinho (LMP@isep.ipp.pt)

- ▶ **Labs**

- ▶ Cláudio Maia (CRR@isep.ipp.pt)

Additional Info

- ▶ Questions about the contents or the functioning of the curricular unit should be sent to the responsible by email
- ▶ Doubts about specific aspects of the subject taught should be raised with the teachers (T, PL as appropriate)
 - ▶ Use email to schedule office hours when needed
- ▶ All documentation and dissemination of the curricular unit will be through Moodle
 - ▶ It is assumed that all students use the ISEP email system